EXP 27 – Best First Search Algorithm

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Aim

To write a Prolog program that implements the Best First Search algorithm using heuristic values to guide the search from a start node to a goal node.

Algorithm

1. Define edges between nodes using edge(Node1, Node2).
2. Assign heuristic values to each node using heuristic(Node, Value).
3. If the current node is the goal, return the path.
4. Otherwise:
    - Find all neighboring nodes.
    - Pair each neighbor with its heuristic value.
    - Sort the pairs based on heuristic values (best first).
    - Choose the node with the lowest heuristic value.
    - Recursively continue the search from that node.
5. Build the final path by accumulating visited nodes.

Source Code

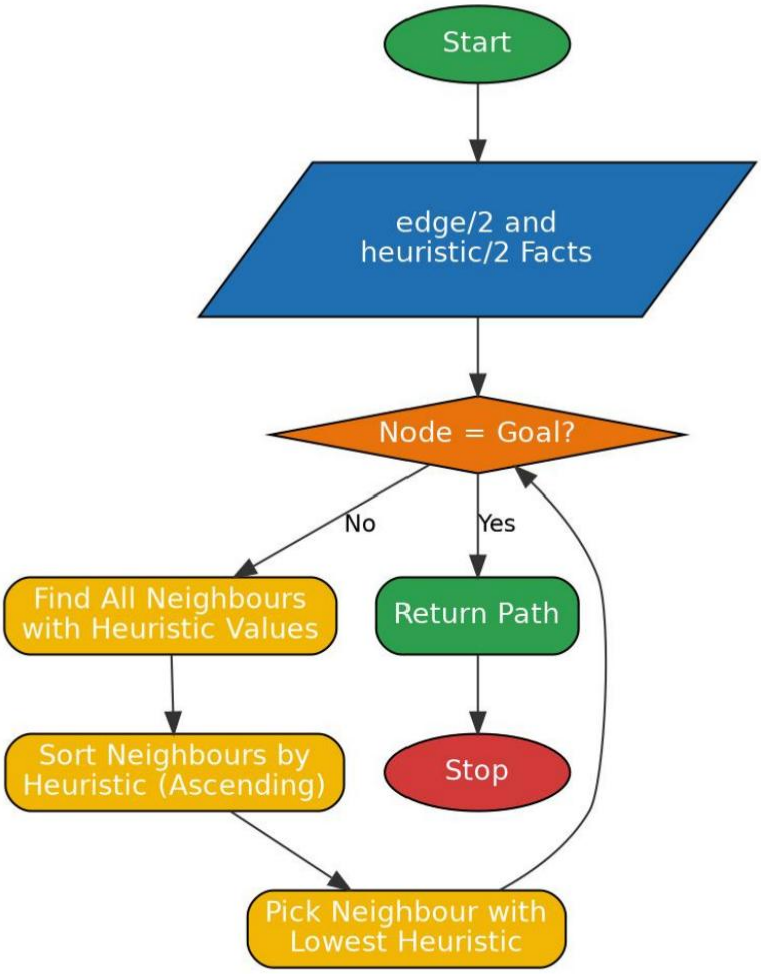
prolog

```
edge(a, b).
edge(a, c).
edge(b, d).
edge(b, e).
edge(c, f).
edge(e, g).
edge(f, g).
```

```prolog
heuristic(a, 10).
heuristic(b, 8).
heuristic(c, 5).
heuristic(d, 7).
heuristic(e, 3).
heuristic(f, 4).
heuristic(g, 0).

best_first(Goal, Goal, [Goal]) :- !.
best_first(Node, Goal, [Node | Path]) :-
    findall(H-Next, (edge(Node, Next), heuristic(Next, H)), Pairs),
    keysort(Pairs, Sorted),
    member(_-Next, Sorted),
    best_first(Next, Goal, Path).
```

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [bestfirst].
true.

?- best_first(a, g, Path).
Path = [a, c, f, g] .
```

Result

The program successfully performs Best First Search by selecting the next node based on the lowest heuristic value. It finds an efficient path from the start node to the goal node using heuristic-guided exploration.